
Exercise 3

1 Pre-set

1. Consider the system

$$x_{k+1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x_k.$$

If $x \in \mathcal{X}$, $\mathcal{X} = \{x \mid \begin{bmatrix} 1 & 1 \\ -1 & -1 \\ 1 & 0 \\ -1 & 0 \end{bmatrix} x \leq \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\}$, sketch the set $\mathcal{X}$ and pre-set of $\mathcal{X}$. Compute the successor state of a few points in $\text{pre}(\mathcal{X})$ to convince yourself it is correct.

2. Consider the system $x_{k+1} = 0.5x_k + u_k$. Calculate and plot the pre-set if $x \in [-2, 2]$ and $u \in [-1, 1]$.

2 Invariant set

1. Consider the system $x_{k+1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x_k$ and the constraint $\{x \mid \begin{bmatrix} 1 & 1 \\ -1 & -1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix} x \leq \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\}$. Calculate and plot the maximum invariant set.
2. Consider the nonlinear system $x_{k+1} = x_k^2 + u_k$, where $x \in [-2, 2]$, $u \in [-1, 1]$. Calculate the maximum control invariant set. (Hint: Consider a sequence, $a_{k+1} = \sqrt{a_k + 1}$, if $a_0 \geq \frac{1+\sqrt{5}}{2} \approx 1.618$, then $a_{k+1} \leq a_k$, and $\lim_{k \rightarrow \infty} a_k = \frac{1+\sqrt{5}}{2}$)
3. Given a system $x_{k+1} = f(x_k)$, which has a Lyapunov function V . (Recall a Lyapunov function has the property $V(f(x_k)) \leq V(x_k)$). Prove that the set $\{x \mid V(x) \leq \alpha\}$ is invariant for all $\alpha \geq 0$.